

Supplementary Materials

From Surface Deformation to Permafrost Process Inference: A Systematic Review of InSAR Applications, Validation, and Quantitative Evidence

Supplementary Note S1. Analytical populations and screening logic

The review distinguishes retrieval, screening, bibliometric, and quantitative-evidence populations because they serve different inferential purposes. The broad search yielded 409 Web of Science records. Restriction to SCIE/SSCI Articles and Reviews produced a 334-publication formal screening library. Topic and scope screening retained 262 publications in the core analytical corpus, comprising 260 empirical Articles and two focused Reviews. Seventy-two publications from the formal library were outside the core eligibility criteria; 24 were retained as contextual or boundary literature for narrative interpretation but were excluded from denominator-based bibliometric analyses.

The structured quantitative evidence set is nested within the core corpus. It contains 175 publications contributing 372 coherent scientific results and 727 companion quantitative metrics. Publication, result, and metric counts are therefore not interchangeable.

Table S1. Analytical populations and their primary roles in the review.

Population	N	Primary analytical role
Broad topic-query output	409 records	Sensitivity-oriented retrieval set
Formal SCIE/SSCI Article/Review library	334 publications	Standardized topic/scope screening denominator
Core analytical corpus	262 publications	Bibliometric and thematic denominator
Empirical core Articles	260	Population assessed for quantitative scientific evidence
Focused Reviews	2	Methodological/application context; excluded from empirical result counts
Formal-library records outside the core	72	Excluded from core denominator-based analyses

Population	N	Primary analytical role
Contextual/boundary set	24	Narrative use only; subset of the 72 outside-core publications
Structured quantitative evidence set	175 publications	Publications contributing at least one coherent scientific result
Coherent scientific results	372	Primary unit of quantitative evidence synthesis
Companion quantitative metrics	727	Dependent descriptors attached to coherent results

Supplementary Note S2. Sensitivity re-screening for quantitative evidence

The 260 empirical core Articles were screened for quantitative information relevant to deformation magnitude, validation, environmental association, physical inversion, cross-sensor or cross-method comparison, and model or algorithm performance. Indexed metadata can underrepresent quantitative content that appears only in full text, figures, tables, or supplementary materials. A sensitivity re-screen was therefore applied to records initially classified as lacking an explicit quantitative signal.

The re-screen identified additional numerical cues in most of these records and generated a focused set of 100 publications requiring completion review. Of these, 90 contained result-level quantitative targets, nine were descriptive-only for the purposes of the structured evidence database, and one contained no relevant quantitative InSAR result. This sensitivity step was used to reduce false-negative extraction rather than to change the final 262-publication core scope.

Supplementary Note S3. Definition of a coherent scientific result

The extraction unit was a coherent scientific result rather than an individual number. One Result_ID corresponds to one scientific analysis, comparison, model evaluation, association, or physical inference. Multiple statistics describing the same analysis are stored as companion metrics.

Examples include:

- RMSE, MAE, bias, and coefficient of determination from the same validation model: one Result_ID with multiple metrics;
- several statistics describing one cross-sensor comparison: one Result_ID;

- separate analyses using the same publication but answering different scientific questions: separate Result_IDs;
- repeated publications using the same underlying radar or field dataset: separate publications and results, but linked through dataset-dependence metadata.

The following rules were applied consistently:

1. R^2 was not converted to Pearson r .
2. Narrative direction was not used to infer an unreported correlation sign.
3. A p-value alone was not treated as an independent result.
4. Pixels, mapped objects, interferograms, scene counts, or study-area size were not treated as analytic sample size unless the original analysis explicitly defined them as the relevant independent denominator.
5. Reviewer-derived quantities were labelled separately from author-reported values.
6. Conflicting source labels or values were preserved with an explicit note rather than silently reconciled.

Supplementary Figure S1. Thematic evolution of the core corpus

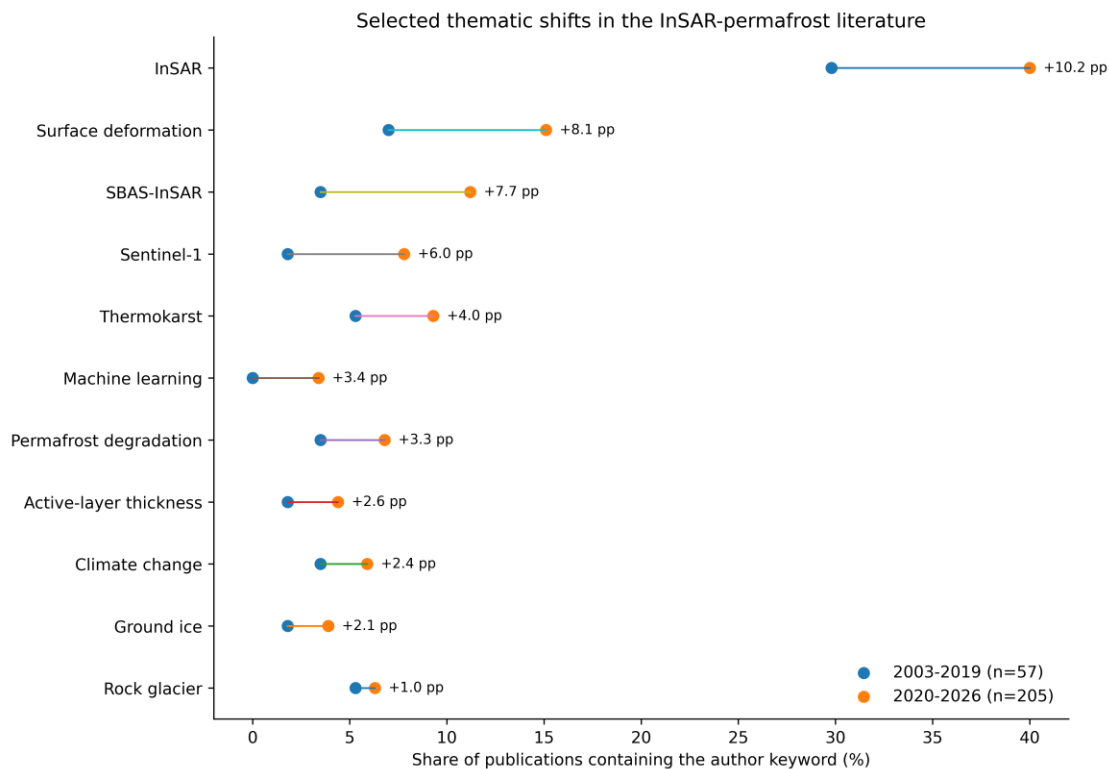


Figure S1. Change in author-keyword prevalence between 2003-2019 ($n = 57$ core publications) and 2020-2026 ($n = 205$). Changes are expressed in percentage points (pp), calculated as the prevalence in 2020-2026 minus the prevalence in 2003-2019.

The analysis is descriptive and indicates changes in research emphasis rather than statistical significance.

Supplementary Figure S2. Country affiliation of the core corpus

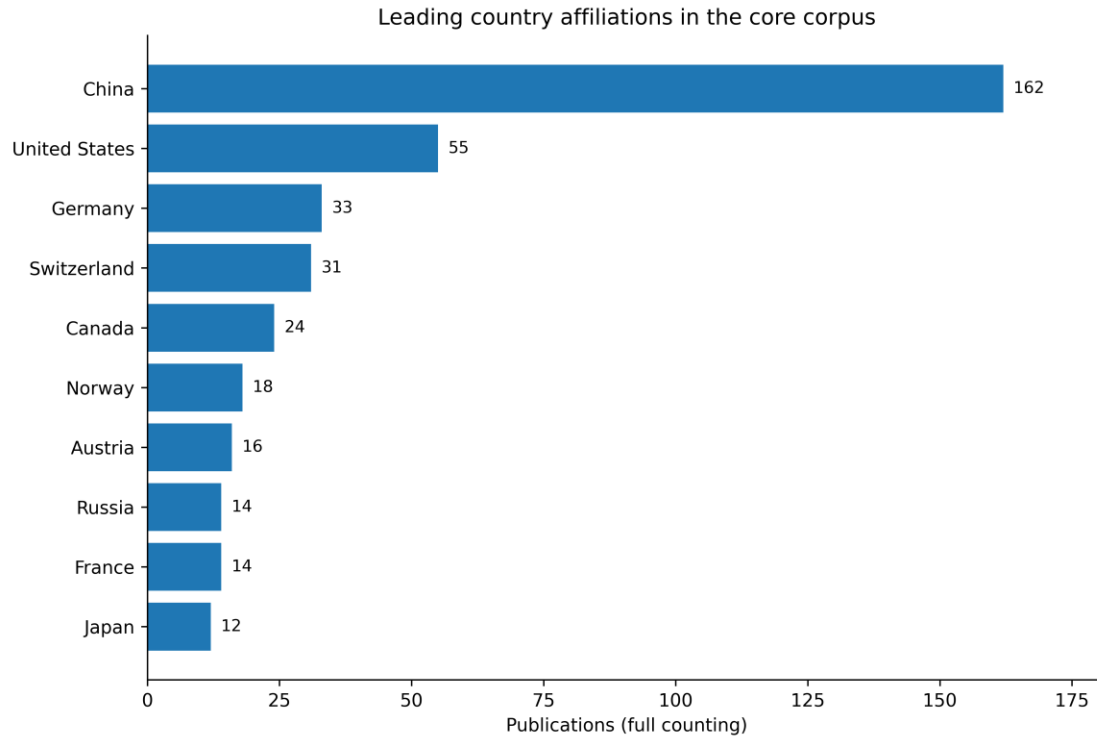


Figure S2. Country participation based on author affiliations using full counting. Multinational publications contribute once to each represented country; totals therefore exceed 262. Affiliation counts indicate research participation and should not be interpreted as geographic study-site coverage.

Supplementary Note S4. Evidence-family definitions and counts

Table S2. Evidence-family definitions and result counts.

Evidence family	Definition	Publications	Coherent results
Descriptive deformation/quality outcome	Deformation magnitude, rate, amplitude, distribution, quality, or event response	89	117
Model/algorithm performance	Prediction, classification, retrieval, calibration, processing, or	57	100

Evidence family	Definition	Publications	Coherent results
Environmental association	forecasting performance Statistical relation between deformation/derived product and an environmental predictor	42	61
ALT/active-layer inference	Estimation, inversion, uncertainty, or validation of active-layer properties	21	29
Ground-ice/excess-ice or hydrological inference	Physical inference concerning ground ice, meltwater, storage, or hydrological contribution	20	25
Cross-sensor/cross-method agreement	Comparison among sensors, geometries, processing methods, or remote-sensing products	15	22
External field/GNSS/levelling validation	Direct comparison with an independent or quasi-independent field/geodetic reference	14	17
Unclassified/boundary evidence	Result not assigned confidently to a principal family	1	1

The single boundary result was retained explicitly rather than forced into a family whose scientific purpose it did not match.

Supplementary Note S5. Reporting completeness

“Explicit analytic n” required a defensible result-specific denominator. Counts of pixels, scenes, mapped landforms, or study sites were not automatically interpreted as analytic sample size. “Usable uncertainty” required a quantity that could inform interpretation or weighting, such as SD, SE, confidence interval, variance, an explicitly defined validation-error metric, or an uncertainty/error bound.

Table S3. Reporting completeness by evidence family.

Evidence family	Results	Explicit analytic n	%	Usable uncertainty	%
Overall	372	46	12.4	126	33.9
Descriptive deformation/quality	117	1	0.9	27	23.1
ALT/active layer	29	6	20.7	13	44.8
External field/GNSS/levelling validation	17	6	35.3	9	52.9
Cross-sensor/cross-method	22	5	22.7	2	9.1
Environmental association	61	4	6.6	5	8.2
Ground ice/hydrology	25	6	24.0	3	12.0
Model/algorithm performance	100	18	18.0	67	67.0

These indicators are synthesis-readiness descriptors rather than conventional risk-of-bias scores.

Supplementary Note S6. ALT validation typology

ALT validation was stratified by reference type and independence because reported statistics estimate different quantities.

Table S4. Representative ALT validation designs and their implications for synthesis.

Study	Reference/validation design	Quantitative information retained	Interpretation for synthesis
Liu et al. 2012	Six usable ground sites; uncertainty-normalized agreement	5/6 sites within reported reference uncertainty	Custom r^2 is an uncertainty-normalized discrepancy statistic, not conventional R^2
Schaefer et al. 2015	ReSALT field validation	ALT validation metrics retained in the source evidence family	ALT-specific validation; do not mix metric definitions without common estimand
Wang et al. 2018	Two Beiluhe field sites	3.1 vs 2.8 m; 2.0 vs 1.5 m	Small site-level comparison; sites also informed model construction
Peng et al. 2023	Seven effective paired field sites per model	Mean errors 1.24/0.47 m; median errors 1.15/0.28 m; internal label conflict for 1.48/0.67 m	Preserve source labels and reviewer-derived checks separately
Chen et al. 2023, PDO Part II	15 CALM sites plus GPR	CALM: RMSE ~15 cm, bias -7 cm; GPR: RMSE 18 cm, bias -10 cm	Stronger external validation; CALM and GPR represent different supports
Garestier et al. 2023	21 retained probing locations	Spatial/categorical corroboration	No standard RMSE, MAE, bias,

Study	Reference/validation design	Quantitative information retained	Interpretation for synthesis or correlation for pooling
Howard et al. 2022	100 sampled locations versus GIPL model	$R^2 = 0.6967$, RMSE 5.20 m, relative RMSE 22.08%	Model-reference comparison, not independent field validation

Peng et al. 2023 metric-label inconsistency

The field-comparison table provides seven valid paired ALT observations for each inversion model after missing values are excluded. Recalculation from the printed paired values yields RMSE values numerically matching the values labelled “standard deviation” in the Results (approximately 1.48 m and 0.67 m). The paper also reports mean errors of 1.24 m and 0.47 m, while the Conclusion refers to 0.67 m as a mean error. The evidence database therefore retains the author-reported labels and separately records the reviewer-derived RMSE check; the inconsistency is not silently corrected.

Liu et al. 2012 custom agreement statistic

The paper defines a site-level quantity equivalent to a squared discrepancy normalized by reference uncertainty. It is not the conventional coefficient of determination and was therefore not converted to Pearson correlation or Fisher-z.

Supplementary Note S7. Confirmed cross-publication dataset dependence

Two cross-publication shared-data groups were explicitly confirmed.

1. **Alaskan North Slope ERS 1992-2000.** The 2012 ALT study explicitly states that it used the same SAR data and a similar processing approach as the earlier 2010 deformation study.
2. **2017 ABoVE airborne campaign.** PDO Part II uses L-band seasonal-subsidence products processed within the PDO Part I framework and the same broader airborne campaign.

The publications remain distinct scientific contributions because they answer different questions, but they should not be treated as fully independent replications of the underlying observations.

Supplementary Note S8. Quantitative-synthesis readiness rules

Formal pooling required all of the following:

1. a common scientific estimand;
2. comparable outcome and predictor/comparator definitions;

3. independent replication or an explicit dependence model;
4. defensible analytic sample size;
5. usable sampling uncertainty or an error structure appropriate for weighting.

Compatible metric definition was evaluated within the common-estimand and compatible-outcome criteria rather than as a separate sixth condition. No current evidence family met all criteria. A global pooled estimate was therefore not calculated.

Pearson r would be eligible for Fisher- z transformation only when a genuine Pearson correlation and a defensible independent n were available:

$$z = \frac{1}{2} \ln \left(\frac{1+r}{1-r} \right), \quad SE_z = \frac{1}{\sqrt{n-3}}.$$

This transformation was not applied to R^2 , Spearman ρ , GeoDetector q , custom agreement statistics, or pixel counts.

Supplementary Note S9. Minimum synthesis-ready reporting framework

Table S5. Minimum information recommended for synthesis-ready reporting.

Domain	Minimum information
SAR acquisition and geometry	Sensor, band, polarization, dates, orbit, incidence angle, acquisition count, gaps, exclusions
Interferometric processing	Network rules, coherence threshold, unwrapping, DEM, atmospheric/ionospheric treatment, filtering, reference, masks
Displacement estimand	LOS or decomposed component, sign convention, seasonal/cumulative/rate definition, temporal model, time window
Permafrost context	Permafrost setting, land cover, terrain, ground ice, soil moisture, snow, hydrology, disturbance history
External validation	Reference method and uncertainty, paired n , independence, footprint/time matching, bias, absolute error, association/agreement
Predictive modelling	Target definition, split strategy, spatial/temporal independence, external validation, uncertainty in target/reference data

Domain	Minimum information
Dependence and reproducibility	Shared dataset/site identifiers, acquisition provenance, code/data availability, machine-readable result tables

Supplementary Note S10. Record-level cautions retained in the evidence database

A small number of result-level records retain caution flags because the original publications contain metric-definition ambiguity, incomplete analytic-denominator information, or internal inconsistencies between abstract, Results, tables, or conclusions. These records remain in the evidence database because the underlying scientific result is relevant, but ambiguous values are not promoted to standardized effect sizes. The complete record-level notes are provided in the accompanying audit file.